THE m_3–$B^{(2)}$ SAGA: CY-A$_3$ AS ∞-CATEGORICAL PROOF VIA COSTELLO TCFT

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Abstract. SCAFFOLD STUB. The CY-A$_3$ claim $\mathrm{Obs}_{A_\infty} = 0$ holds as an ∞-categorical identity on the inf-cat derived centre (scope: derived, not chain-level per-k). Chain-level, the individual anticommutators $\{b_k, B^{(2)}\}$ do *not* all vanish; the *total* sum $\sum_k \{b_k, B^{(2)}\}$ vanishes by Costello's TCFT cyclic invariance. This is a paradigmatic instance of the principle "chain-level ≠ rational": E_3 collapses under formality, so an obstruction that is nonzero at chain level becomes a boundary (Q-exact) after formality transport. The saga resolves the per-k discrepancy and gives the first ∞-categorical CY-A proof.

Primary theorems

- CY-A$_3$: $\mathrm{Obs}_{A_\infty} = 0$ on the inf-cat derived centre: .
- Individual $\{b_k, B^{(2)}\} \neq 0$ (chain-level): (falsifies naive per-k identification).
- $\sum_k \{b_k, B^{(2)}\} = 0$ (total sum) via Costello TCFT cyclic invariance: .
- Chain-level m_3 need not be a derivation after formality: (chain-level vs rational scope).

Chapter inputs (pointers)

```
\input{../chapters/theory/m3_b2_obstruction.tex}
\input{../chapters/theory/cyclic_ainf.tex}
\input{../chapters/theory/hochschild_calculus.tex}
\input{../chapters/theory/drinfeld_center.tex}
```

Cross-volume dependencies

- Vol I Theorem $A^{\infty,2}$ properad (`thm:A-infinity-2`) – provides the Francis–Gaitsgory ambient in which inf-categorical CY-A$_3$ is stated.
- Vol II `thm:chiral-higher-deligne` – the E_3 target operad into which CY-A$_3$ maps.

Discipline anchors

Chain-level vs rational invariance; total cyclic sum vs per-k chain-level anticommutators; independent verification of every multi-step structure (arrow-by-arrow).

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